

Reproduction of pytorch/examples: Transformer word-level language model on WikiText-2

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Independent reproduction · Benchmark or leaderboard entry · Natural language processing

Abstract

Trained the word_language_model example from the pinned commit unmodified: a Transformer language model (default size: 200-dim embeddings, 2 layers, 2 heads, dropout 0.2) on the WikiText-2 copy shipped in the repo, 20 epochs, initial learning rate 5 with the script annealing schedule, seed 1111, on the GPU (--accel, which replaces the older --cuda flag). Validation loss and perplexity per epoch and the final test loss and perplexity are parsed from the script output. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Transformer LM WikiText-2	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: Transformer LM WikiText-2

Run	Metric	Final value	Step
1	valid_loss	5.07	20
1	valid_ppl	159.49	20

Run	Inline claim	Value	Phase
1	test_loss	5	
1	test_ppl	148.78	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	786.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	5d1c96253b982df87e352be3841a7870905e887991d904db0d1f1556d4d207aa	6.7 MB
r1.kvbundle.zip	a213f90f61722cf9b42ceda507608b90af82eb65cf5332f439b53f385eae1b2a	213 KB

5 Links and licence

Reproduction of: <https://github.com/pytorch/examples/tree/acc295dc7b90714f1bf47f06004fc19a7fe235c4>

Record: CC-BY 4.0. Code: BSD-3-Clause, as declared by the authors. Tags: language-model, transformer, wikitext-2, pytorch.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Transformer LM WikiText-2), kv:2610.00009v1/r1. Data hash 349ed2af849a2c67dbe68b1d153cedcb4523a1c5ebdd45cc3f17dd7d64e7737e.
<https://kveritas.org/records/2610.00009v1/r1.pdf>